Mastering Gemini Gems

Building Your Digital Teaching Fellows

Persistent · Personalized · Pedagogically Aligned

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GEMINI × PEDAGOGY

Why Use Gems? — Beyond One-Off Prompts

INTRODUCTION

A Gem is a ***persistent assistant*** — pre-conditioned with your pedagogy, syllabus, and teaching style. No re-prompting. No starting over.

WITHOUT A GEM

- ✗ Re-paste your syllabus every session
- ✗ Re-explain your grading philosophy
- ✗ Re-set the persona each time you open a new chat
- ✗ Inconsistent tone and guardrails

WITH A GEM

- ✓ Opens knowing your course context
- ✓ Maintains the persona you designed
- ✓ Enforces your pedagogical guardrails automatically
- ✓ Scales across every student interaction

Technical Setup — The 5-Step Process

HOW-TO GUIDE

1 ACCESS

Open Gemini and click "Explore Gems" in the left sidebar.

2 DEFINE

Click "New Gem" and give it a clear name — e.g., "Socratic CS Mentor."

3 INSTRUCT

Write your persona in the Instructions box. Use the pencil "Rewrite with AI" icon to polish rough notes into a professional prompt.

4 KNOWLEDGE

Upload up to 10 files — PDFs, syllabi, rubrics — to give the Gem specific, citable context.

5 TEST & SAVE

Preview the Gem in the test window, verify the tone and guardrails, then save it to your sidebar for instant access.

The Persona Framework — The Four Pillars

PERSONA FRAMEWORK

Every great Gem instruction set contains these four elements — build all four or the Gem will drift.

ROLE

Who is the AI?

Define the persona explicitly.

Example: "You are a picky peer reviewer for NSF proposals."

KNOWLEDGE

What does it know?

Point to uploaded files or domains.

Example: "Reference the uploaded 2026 ABET criteria."

STYLE/TONE

How does it speak?

Set register and vocabulary rules.

Example: "Supportive but direct; never use jargon without defining it."

GUARDRAILS

What must it never do?

Hard limits that prevent shortcuts.

Example: "Never give the final answer; only provide guiding hints."

All four pillars work together — Role without Guardrails will hallucinate authority; Guardrails without Role will feel robotic.

Example Personas — Three Ready-to-Use Gems

FACULTY EXAMPLES

Student-Facing

SOCRATIC STUDY PARTNER

Guide students through difficult concepts without doing the work for them.

"You are a Socratic tutor for Intro to Databases. Guide students through ACID properties via questioning. If a student asks for a definition, ask a question that forces them to deduce the answer from a bank transfer or gaming loot-drop scenario."

Research-Facing

PROPOSAL PRE-REVIEWER

Act as a critical reviewer for NSF/NIH grant proposals.

"You are a seasoned NSF Program Director. Review the project summary for intellectual merit and broader impacts. Be brutally honest about gaps in the Responsible AI section. Flag fluff and suggest specific, data-driven alternatives."

Pedagogy-Facing

INCLUSIVE CURRICULUM ARCHITECT

Help faculty rewrite assignments for better accessibility and UDL compliance.

"You are an expert in Universal Design for Learning. Analyze the uploaded assignment and suggest three ways to improve accessibility for neurodivergent students or English language learners. Check specifically for clarity in the grading rubric."

The Feynman Engine — Why Gems Work as Socratic Mentors

1 Choose a Concept

Pick what the student must genuinely understand.

2 Teach It Simply

Explain it in plain language — as if to a 12-year-old.

3 Find the Gap

Where do you stumble? That is where real learning lives.

4 Refine & Simplify

Return to the source, resolve the gap, and re-explain.

A Socratic Gem embeds this loop: it refuses to give answers (step 3), forces explanation (step 2), and redirects students back to source material (step 4) — automatically, every session.

Workshop Tip: Gems stay within the Google Workspace ecosystem — they can interact with Docs and Gmail. For Education accounts, data is protected and ***not used to train global models.***